Gastroesophageal Reflux Disease *Prodigy Anesthesia*

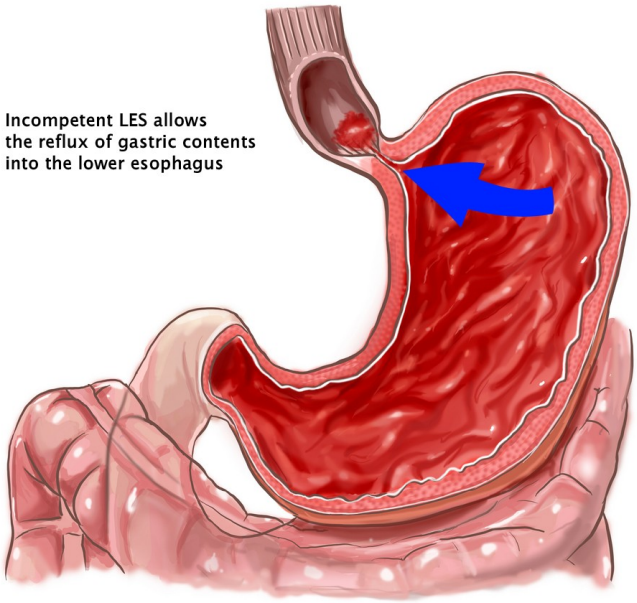
Gastroesophageal reflux disease (GERD) consists of the movement of the contents of the stomach upward through the lower esophageal sphincter (LES) into the esophagus. The three mechanisms that help prevent this from occurring are the pressure exerted by the LES, the crural diaphragm, and the location of the gastroesophageal junction below the diaphragm. The LES normally maintains a resting pressure of about 30 mmHg, which prevents the reflux of stomach contents. In patients with GERD, however, the LES pressure is usually less than half the normal value. The significance of GERD to anesthesia is that it presents an increased risk that gastric contents could move far enough up the esophagus and into the oropharynx and be aspirated into the lungs on inspiration.

The principal symptoms of GERD is retrosternal pain. The chronic aspiration of small amounts of gastric contents during sleep may present as morning hoarseness, chronic cough, laryngitis, pharyngitis, aspiration pneumonia, pulmonary fibrosis, or asthma. Esophageal strictures can also occur due to chronic mucosal damage.

Factors known to increase the risk of aspiration include: lithotomy position, pregnancy, smoking (despite the fact that smoking reduces the incidence of nausea), diabetes mellitus, obesity, increased intra-abdominal pressure, emergency or urgent surgery, and difficult airway.

Drugs that are known to decrease the LES and potentially increase the risk for gastric aspiration include: anticholinergics, dopamine, thiopental, opioids, propofol, tricyclic antidepressants, and sodium nitroprusside.

Drugs that are known to increase the LES and reduce the risk of aspiration include: metoclopramide, prochlorperazine, edrophonium, neostigmine, pancuronium, alpha-adrenergic agents, and antacids. Succinylcholine increase the LES, but also increases gastric pressure during fasciculations, so the overall barrier pressure is left unchanged.



If the timing and nature of the surgery permit, medications to prevent reflux or minimize the acidity of the gastric contents may be given. In general, proton-pump inhibitors are most effective if given the evening prior to surgery and again on the morning of surgery. Cimetidine and ranitidine are effective at reducing the secretion of gastric acid as well as reducing the acidity of the gastric acid that is being secreted. The onset of these effects with cimetidine is about 60-90 minutes and it will last approximately 3 hours. Sodium citrate is a nonparticulate antacid that is effective at reducing the acidity of the gastric contents quickly and is often used prior to urgent surgeries in which the timing of the surgery takes precedent over the patient's NPO status. Tracheal intubation is considered the gold standard method of airway management in the anesthetized patient at risk for aspiration. During induction, cricoid pressure (compression of the esophagus between the cervical vertebrae and the cricoid cartilage) should be applied until the endotracheal airway is established and verified.

Hines RL, Marschall KE. Stoeltings Anesthesia & Co-Existing Diseases. 6th ed. New York, NY: Churchill-Livingston; 2012: 288-289.

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